Data notes: data centers in the US production and investment networks

P&P project (spin-off of RbcProdNet)

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Scope and status (read first)

- **Question.** Can we put data centers, and their energy use, into the 37-sector (41 after the energy split of `data_notes.tex`) production and investment networks, and use the node to study the 2023– investment boom? This note collects what the official statistics identify, what outside sources add, and how the node would be built. It does not decide how the boom is modelled; Section 5 lists the options and what the literature has done.
- **Two objects, not one.** A data center is (i) a *capital good*: a structure plus servers/GPUs, networking, electrical and cooling equipment, owned by whoever operates it; and (ii) a *producing establishment* selling computing, storage and hosting services. The investment boom is about (i); the energy use is about (ii), because electricity is an operating input of the establishment, not of the building. The network needs both: a data center *node* whose capital is the data center *bundle* (a column of the investment network) and whose main intermediate input is electricity (a column of the IO table).
- **Where it is in the data: not in Real estate.** Data center establishments are NAICS 518210 (colocation, cloud infrastructure, hosting; the 2022 NAICS manual sends colocation explicitly to 518210, not to lessors 531120). They sit in BEA industry 514 "Data processing, internet publishing, and other information services", i.e. inside our sector 26 `Info`. Hyperscaler data centers are own-account capital of their parents' industries (Amazon/AWS, Alphabet, Meta in 514; Microsoft and Oracle in 511 software publishers); colocation landlords Equinix and Digital Realty are REITs for tax and SEC purposes but 518210 by activity; enterprise data centers are own-account capital of every industry. The split is therefore of `Info`, with a secondary adjustment to `Real_estate`, whose landlords buy tenants' electricity (Section 1).
- **What BEA now publishes.** Since the 25 September 2025 annual update, NIPA table family 5.4.x has a "Data centers" line under office structures (line 5 of 5.4.5U, series LA001282), quarterly from 2020Q1, sourced from the Census construction-spending data center category (published since May 2024, monthly back to January 2014). This is the *structures* part only: \$20bn in 2023, \$35bn in 2024, \$50bn in 2025 (Census annual), about \$70bn at annual rate by mid-2026. There is no data center asset type in the fixed-asset accounts by industry (last update 26 Sept 2025), no data center industry in the IO accounts, and no data center equipment series. The Fed (Brandsaas et al. 2025) imputes equipment as computers investment above its 2015–2022 trend and finds structures $\approx 30\%$, equipment $\approx 70\%$, with 80% of the equipment imported (Section 1.3).

- **Energy.** Data centers run on grid electricity: 176 TWh in 2023 (4.4% of US electricity), 192 TWh in 2024 (4.7%), projected 325–580 TWh in 2028 and 520–840 TWh in 2030 (LBNL). On-site gas generation is about 2 GW online in mid-2026 against roughly 50 GW of data center load, with 80 GW announced; nuclear and renewable PPAs are contracts, not physical supply. In an IO table the node buys from the electric power sector, and the fuel behind that electricity is the generation mix of the grids where data centers sit (Virginia and Ohio about 60% gas, against 43% for the US). Electricity is about 7% of the all-in annual cost of an AI data center, 25–30% of the cost of the facility excluding IT, 15–27% of colocation revenue, and about 1% of the gross output of industry 514 (Sections 2–3).
- **Verdict.** Feasible in two months as a *single* data center node carved out of `Info`, defined by function (all US data center capacity regardless of owner) with a 2023 or 2024 snapshot steady state: gross output from capacity times engineering cost per MW, IO column from the 2017 detailed Use table for 518200 re-weighted to an engineering cost structure, investment column from BEA fixed assets for industry 5140 plus the NIPA data center structures line, sales row to `Info`, `Prof_Tech`, `Finance`, `Retail` and consumption. Not feasible officially: a time series of data center output or TFP (industry 514 is payment processing and internet publishing before 2015), and a separation of hyperscaler own-use capacity from their parents' other capital. Two modelling issues must be settled before the data work: how the boom is triggered (news about future productivity is the closest fit to the timing) and how to treat imported IT equipment in a closed-economy model in which sector 9 `Computers` would otherwise produce it (Section 4). The current answer to the second is "US+Nvidia": sector 9 augmented with Nvidia's US-installed data-center revenue as a factoryless producer, with the foundry, memory and assembly content as an imported non-produced input (Section 6 and `data_center_nowcast_plan.md`).

1 Where data centers are in the official statistics

1.1 Classification

- **NAICS 2017 518210** "Data processing, hosting, and related services": web and application hosting, data storage, and data processing (payroll, payment, and record processing). **NAICS 2022 518210** "Computing infrastructure providers, data processing, web hosting, and related services" adds IaaS/PaaS, cloud computing and storage, colocation, and cryptocurrency mining explicitly; it moves on-site facilities management to 541513 and payroll to 541214; internet publishing (519130) goes to 513 and 516210 and web search to 519290. The 2022 manual entry for 531120 (lessors of nonresidential buildings) states that colocation is classified in 518210. Economic Census product lines for 518210 include "colocation services" and "cloud computing and cloud storage". `https://www.census.gov/naics/reference_files_tools/2022_NAICS_Manual.pdf`
- **BEA industry codes.** Detailed benchmark (about 400 industries, 2007/2012/2017): 518200 data processing and hosting; 519130 internet publishing and web search; 511200 software publishers; 531ORE other real estate. Summary (71 industries, annual 1997–2023): 514 = 518 + 519; 511; ORE. Our sector 26 `Info` = 511 + 512 + 513 + 514. BEA's industry accounts remain on NAICS 2017 through the 2023 comprehensive update (the 2022 NAICS 516 split is a stated conversion problem); the 2023 update also moved equity REITs from finance into real estate.
- **Company vs establishment.** SEC SIC codes: Equinix and Digital Realty 6798 (REIT); Amazon 5961 (retail); Microsoft, Oracle, CoreWeave 7372 (software); Alphabet, Meta 7370.

BEA's fixed assets by industry are on a company basis, so own-account servers bought by Microsoft are investment of 5110 and by Alphabet, Meta and Amazon-AWS of 5140 (Amazon's fulfilment capital is 4540/retail). The IO accounts are establishment-based, so a hyperscaler's data center establishment is 518210 whatever the parent's SIC. Enterprise data centers never appear as data centers anywhere.

- **Ownership of US capacity.** FERC's 2025 State of the Markets puts about 50 GW of data center capacity in service at end-2025 (24% annual growth since 2020). CBRE counts 9.4 GW of wholesale colocation inventory in eight primary markets at end-2025 (Northern Virginia 4.0 GW) and 6.0 GW under construction, excluding hyperscaler-owned campuses. Synergy's global split at end-2025: hyperscale 48% of capacity (60% owned, 40% leased), other colocation 20%, enterprise on-premises 32%; the US is more hyperscale-owned than that average. Applied to 50 GW: hyperscale-owned $\approx$ 14 GW, hyperscale-leased $\approx$ 10, other colocation $\approx$ 10, enterprise $\approx$ 16 (illustrative). LBNL: 74% of servers were in hyperscale and colocation facilities in 2023, 85% by 2028, and more than 90% of server energy.

1.2 Input-output accounts: what industry 518200 buys and sells

Extracted from `IOUse_After_Redefinitions_PRO_2017_Detail.xlsx` and the 71-industry annual Use tables (BEA `AllTablesIO.zip`, `https://apps.bea.gov/industry/iTables%20Static%20Files/AllTablesIO.zip`); producers' prices, $ millions.

Industry 518200, detailed Use table	2007	2012	2017
Gross output (intermediates + value added)	87,626	117,656	224,377
Total intermediate inputs	34,142	58,121	91,991
Compensation of employees	25,671	18,803	43,130
Gross operating surplus	26,493	39,104	86,201
Purchases of electric power (221100)	556	881	2,215
as % of gross output	0.63	0.75	0.99
Purchases of natural gas distribution (221200)	–	–	5
Purchases of other real estate (531ORE)	1,799	3,050	7,541
Purchases of wired telecom (517110)	1,250	1,256	1,284
Purchases of software publishers (511200)	–	–	1,220
Purchases of computer systems design (541512)	–	–	4,060

Largest intermediate inputs of 518200 in 2017: other real estate (7.5bn), advertising (7.0), employment services (6.6), broadcasting (6.1), lessors of intangibles (5.3), management consulting (4.7), computer systems design (4.1); electricity ranks 18th. Value added is 59% of gross output and gross operating surplus 38%, consistent with a capital-intensive industry; but an electricity share of 1% says that in 2017 the industry's output was dominated by activities other than running servers. Census SAS confirms it: purchased electricity of NAICS 518 was $980m in 2017 (0.8% of $127bn of expenses; the series ends in 2017). For comparison, US data center electricity use in 2017 was about 100 TWh ($7–8bn at commercial rates), of which 518200 records 2.2bn, 519130 (Google, Meta) 0.7bn and 511200 (Microsoft) 0.6bn; the rest sits in 531ORE, which buys $36bn of electricity (13% of all intermediate electricity use) because landlords pay tenants' power, and in enterprise data centers in every industry.

Make table 2017. Commodity 518200 (total 248bn) is produced by industry 518200 (215bn) and, as a secondary product, by software publishers 511200 (33bn: Microsoft's cloud); real estate produces none of it, so BEA does not treat colocation as real estate output. Industry 518200 also produces 7.8bn of software.

Who buys 518200 output (2017 row): intermediate use 171bn, personal consumption 32bn. Top buyers: nondepository credit intermediation (23.9bn, 14% of intermediate use: card processing), state and local government (14.2 + 7.8bn), employment services (11.4), depository credit (9.1), travel reservation (4.5), restaurants (4.4), electric power (4.4), motor vehicle dealers (4.0). This is the row of a business-services industry; a data center node built from it inherits payment-processing customers unless the row is re-weighted with cloud revenue by customer industry.

Industry 514, summary Use table ($bn)	1997	2007	2012	2017	2020	2022	2023
Gross output	47.3	126.1	211.0	400.2	551.4	750.0	802.1
Intermediate inputs	13.4	44.5	99.2	159.6	229.3	333.0	319.2
Compensation	27.6	38.3	38.1	91.7	108.1	136.2	156.6
Gross operating surplus	5.7	41.1	71.1	143.4	209.5	272.1	317.3
Purchases of utilities (22)	0.18	0.60	1.22	3.04	5.65	10.33	8.26
as % of gross output	0.4	0.5	0.6	0.8	1.0	1.4	1.0
Sales to PCE	2.8	15.9	28.8	46.7	93.0	125.8	141.4
Other real estate (ORE): utilities purchases	29.0	54.4	55.4	42.4	24.6	40.1	41.5

Industry 514 grew 17-fold in nominal gross output over 1997–2023 and its utilities bill 45-fold, but the utilities share never exceeds 1.4%. The 2023 figure (8.3bn) is 40–60% of what LBNL's 176 TWh implies ($14bn at industrial, $22bn at commercial rates). Census SAS revenue of NAICS 518 alone: $220bn (2020), $271bn (2021), $308bn (2022); QSS about $329bn (2023) and $375bn (2024). BEA's Digital Economy Satellite Account (discontinued; last release December 2023, years 2017–2022) measured "cloud services" gross output at $187bn in 2021 and "internet and data services" at $213bn, with cloud value added up 232% over 2017–2022; the SAS "hosting and IT infrastructure provisioning" product line was $110bn in 2017 and $156bn in 2020.

1.3 Fixed assets: NIPA data center structures and investment by industry

- **NIPA 5.4.5U/5.4.6U**, line 5 "Data centers" (series LA001282 current $, LB001282 chained, IA001282 price), under office within commercial and health care structures, quarterly from 2020Q1, introduced 25 Sept 2025. Source: Census Value Put in Place. Deflator: BEA composite with equal weights on the BLS PPIs for industrial buildings and warehouses (the rest of office keeps the office-building PPI). FRED still shows office as one line ($100.5bn 2024, $102.6bn 2025). `https://www.bea.gov/information-updates-national-economic-accounts-2025`; series register `https://www.bea.gov/sites/default/files/2025-08/NEA-AU25-series-register.txt`.
- **Census C30**, private nonresidential, office: data center; first published with the May 2024 release, monthly back to January 2014; definition "buildings that contain the hardware needed for storing, processing, and transmitting digital information"; racks and servers excluded. Annual value put in place, $bn (`https://www.census.gov/construction/c30/pdf/region.pdf`): 2019 8.5; 2020 9.2; 2021 9.9; 2022 12.6; 2023 20.0; 2024 34.8; 2025 49.7 (South 27.2, Midwest 13.1, West 8.9). 2014–2018 are in the historical Excel files (about $1.6bn annual rate in January 2014). Monthly annual rate: $40bn (April 2025), $51bn (April 2026), about $68–75bn (June–July 2026, +57% y/y; secondary reports of the 1 Sept 2026 release). An industry source's "$162bn of BEA data center investment in 2025" is not this line and should not be used.
- **Equipment in NIPA 5.5.5U**: computers and peripheral equipment (B935RC) $160.3bn (2022), 149.1 (2023), 178.7 (2024), 270.7 (2025); quarterly annual rate 222.8, 253.8, 280.9, 325.4 in 2025 and 384.2, 400.6 in 2026Q1–Q2 (FRED, 4 Sept 2026 vintage; real computers

investment B935RX fell from 356.1 to 348.4 between the two quarters while the nominal series rose). Communication equipment (C275RC) 141.9 (2024), 163.7 (2025). Information processing equipment (Y034RC) 501.1 (2024), 626.6 (2025), 766.5 in 2026Q1. Software 754.8 (2025). Shares of private nonresidential fixed investment ($4,250.8bn, 2025): IPE 14.7%, software 17.8%, together 32.5%. Servers and GPUs are in "computers and peripheral equipment" (fixed-asset types EP1A mainframes, EP1B PCs, EP1G storage, EP1H system integrators); networking in EP20; there is no GPU or data center equipment line.

- **Detailed fixed assets by industry × asset type** (`detailnonres_inv1/stk1/dep1.xlsx`, `https://apps.bea.gov/national/FA2004/Details/Index.htm`, updated 26 Sept 2025, 1901–2024): *no* data center asset type; structures are SOO1 office, SU20 communication, SC02 other commercial, etc. Industry 5140 in 2024, $bn: total investment 282.1, of which equipment 78.8 (mainframes 10.4, PCs 11.0, system integrators 19.5, storage 1.0, communications 18.4, other electrical 4.0, general industrial 1.7), structures 36.9 (office 20.7, communication 12.9), IPP 166.3 (software 133.5). Computers (EP1A–H) are 45.2bn, 26% of the all-industry total (171.4bn); office structures 20.7bn, 21% of the total (99.6bn). Industry 5110 (Microsoft) invests 12.0bn in computers and 4.4bn in communications equipment; real estate 5310 invests 5.5bn in computers and 15.1bn in office structures.

Industry 5140 investment, $bn	2014	2017	2019	2021	2023	2024
Equipment	22.3	37.4	37.4	59.2	70.8	78.8
computers EP1A–H	14.2	19.3	20.3	34.0	37.9	45.2
communications EP20	5.7	10.8	10.0	14.5	18.5	18.4
other electrical EO72	0.6	1.8	1.8	2.6	3.8	4.0
Structures	7.5	9.8	23.4	29.2	36.8	36.9
office SOO1	3.5	4.9	13.2	16.9	20.8	20.7
communication SU20	2.8	4.2	8.4	10.2	12.7	12.9
Intellectual property (software)	36.5	68.8	79.0	118.6	156.4	166.3
Total	66.3	116.0	139.8	207.0	264.0	282.1

- **Net stocks (current cost, 2024)** for 5140: equipment 245bn (communications 84, system integrators 41, PCs 22, mainframes 21, other electrical 14, general industrial 7), structures 304bn (office 171, communication 104), IPP 378bn. Depreciation 2024: equipment 53.2bn (implied rate 22%), structures 7.4bn (2.4%), IPP 131.6bn (35%).
- **Scale check.** Big-4 hyperscaler capex was about $210–230bn in 2024 and $360–410bn in 2025 (Section 5), worldwide and including finance leases and non-data-center capital; BEA records 45bn of computers for 5140 and 12bn for 5110 in 2024, and 179bn of computers for the whole economy. Part of the gap is foreign capex; part is that BEA's allocation of equipment across industries is, in the readme's words, "based on judgmental trends"; part is GPUs bought by CoreWeave/Oracle (industry 5110 or 5415) or leased. The 5140 industry-by-asset detail is usable for *shares* of the bundle, not for its level.
- **Fed measure (Brandsaas, Garcia, Kurtzman, Nichols, Zytek, FEDS 2025-109).** Data center investment = NIPA data center structures + (computers investment − its 2015–2022 linear trend). About $60bn annual rate in 2024Q4, $180bn in 2025Q4, $370bn by 2026Q2 (mean nowcast from Dodge project data). Split 30% structures, 70% equipment (from 2024 Census and BEA data); equipment import share 80%; hence 44% of data center investment is domestic. Contribution to nominal GDP growth 0.17pp (2024), 0.41pp (2025), 0.75–1.1pp (2026) before the import adjustment. Plans: $7bn a year in 2019–22, $300bn in 2024, $900bn in January–September 2025; abandonment 0.33; plan-to-start 11 months; start-to-completion 19 months for a $1bn project. `https://doi.org/10.17016/FEDS.2025.109`
- **GDP by industry**, value added of 518–519 (current $bn): 315.6 (2020), 369.1 (2021), 416.8

(2022), 471.2 (2023), 529.1 (2024); real (2017$) 467 (2022), 539 (2023), 608 (2024). Real output of 518 is deflated with the BLS PPI for 518210, which rises 1.4% a year (2016–2025; matched-model contracts), whereas hedonic cloud prices fell 7–17% a year over 2009–2016 (Byrne, Corrado and Sichel 2018) and list prices have been flat since 2017; measured real output and TFP of the industry are understated.

2 Energy use of data centers

2.1 Electricity consumption

Source	Year	TWh	% of US	Notes
LBNL 2024 (Shehabi et al.)	2014	58	1.5	
	2018	76	1.9	
	2023	176	4.4	servers ≈ 100 TWh, of which GPU servers > 40; storage 16
	2028	325–580	6.7–12.0	74–132 GW at 50% utilisation
LBNL 2025 update (Smith et al., June 2026)	2024	192	4.7	2023–24 revised slightly down
	2028	464		reference
	2030	649 (521–843)	11.8 (9.5–15.3)	AI servers 55% of DC energy; 148 GW
IEA Energy and AI (April 2025)	2024	≈ 187		45% of 415 TWh world; +240 TWh to 2030
EPRI 2026	2024	177–192		2030: 380–790 TWh, 9–17%
EIA AEO2026	2025			servers 7% of commercial electricity; 22–33% by 2050
Grid Strategies 2025	2030			utility peak +166 GW, of which ≈ 90 GW data centers

Denominator (EIA Electric Power Annual Table 2.2): retail sales 3,765 TWh (2014), 3,859 (2018), 3,874 (2023), 3,975 (2024); total end use including direct use 4,011 (2023), 4,110 (2024). Fleet PUE 1.55 (2018) to 1.45 (2024); AI facilities 1.145; infrastructure (cooling, UPS) is 31–36% of data center electricity. Reports: `https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf`; `https://www.osti.gov/servlets/purl/3374245`; `https://www.iea.org/reports/energy-and-ai`; `https://powering-intelligence.epri.com/executive-summary.html`; `https://www.eia.gov/outlooks/aeo/`.

2.2 Kind of energy: physical vs contractual

- **Grid electricity** is at least 95% of physical supply in 2024. No official series separates it.
- **Behind-the-meter gas** (2024–): xAI Colossus, Memphis, about 1.5 GW of turbines operating; Meta Hyperion, Louisiana, three Entergy CCGTs of 2.26 GW (ten plants, > 7 GW announced); OpenAI/Oracle Stargate Abilene, 2.45 GW planned; Crusoe, Microsoft, Amazon projects. Cleanview (mid-2026): about 2 GW online, 2.8–3.2 GW by end-2026, 82 GW announced since 2025. GE Vernova gas turbine backlog 116 GW (20% data center customers), Siemens Energy 69 GW, Mitsubishi 35 GW; Global Energy Monitor counts 378 GW of US gas plants in development. National on-site gas is 1–2% of data center energy today; the announced pipeline would make it a first-order channel from data centers to gas demand by 2028–30. `https://cleanview.co/reports/behind-the-meter-data-centers`

- **Diesel backup**: large installed standby capacity (Virginia alone about 27 GW nameplate) run 10–30 minutes a month for testing; negligible energy. **Fuel cells**: Equinix–Bloom 75 MW operating; negligible.
- **Nuclear PPAs** (contractual, grid-delivered): Microsoft–Constellation Three Mile Island 835 MW from 2028; Meta–Constellation Clinton 1,121 MW from 2027; Amazon–Talen Susquehanna up to 1.9 GW; Google–NextEra Duane Arnold 615 MW from 2029; Google–Kairos 500 MW by 2035; Meta–Vistra/TerraPower/Oklo up to 6.6 GW by 2035. **Renewable PPAs**: the four hyperscalers signed 49% of the 55.9 GW of global corporate clean-energy PPAs in 2025 (BNEF); Equinix reports 96% renewable coverage on a market basis. In the IO table the node buys electricity from 2211 regardless of these contracts.
- **Fuel behind the electricity** (EIA 2024 generation): US gas 43%, nuclear 18%, coal 15%, wind 10.5%, hydro 5.6%, utility solar 5%. Virginia: gas 59%, nuclear 28%, and sales (138 TWh) exceed in-state generation (103) by a third, met by PJM imports; PJM's 2026/27 cleared capacity is 45% gas, 22% coal, 21% nuclear. Texas: gas 52%, nuclear 7%, wind and solar large. Arizona: gas 47%, nuclear 28%. Ohio: gas 60%, nuclear 13%. Georgia: gas 41%, nuclear 34%. Dominion's 2024 IRP: demand +5.5% a year (doubling by 2039, +0.5% without data centers), 5.9 GW of new gas in 2030–36 in every scenario. For the model this means the marginal electricity bought by data centers is more gas-intensive than the national average: the data center node interacts with the fossil-generation share of the electricity split in `data_notes.tex`.

2.3 Electricity bill and price effects

At EIA average prices (commercial 12.6–12.8 cents/kWh, industrial 8.0–8.1 cents, 2023–24), 176 TWh cost \$14–22bn in 2023 and 192 TWh \$16–25bn in 2024; large data centers pay between the two rates. PJM capacity prices went from \$28.92/MW-day (2024/25) to \$269.92 (2025/26; \$444 in Dominion) and the \$329.17 cap (2026/27, uncapped \$388.57); the market monitor attributes \$21.3bn (45%) of the last three auctions' \$47.2bn to data center load, much of it forecast rather than built. CMU/NC State: data centers and crypto raise US generation cost 8% by 2030 and more than 25% in northern Virginia. IMF WP 25/081 (CGE): US electricity prices +8.6% under constrained renewables and transmission. Twenty-four states have large-load tariffs (EEI, August 2026). Harvard ELI (Martin and Peskoe 2025) documents the socialisation of interconnection costs in about fifty rate cases. `https://www.pjm.com/-/media/DotCom/markets-ops/rpm/rpm-auction-info/2026-2027/2026-2027-bra-report.pdf`; `https://www.imf.org/en/publications/wp/issues/2025/04/21/power-hungry-how-ai-will-drive-energy-demand-566304`

3 Cost structure of a data center node

3.1 Evidence

Source	Electricity share and cost structure
Census SAS, NAICS 518	Purchased electricity \$980m in 2017 on \$127bn expenses (0.8%); purchased fuels \$12m (2012). Series discontinued after 2017 (folded into other expenses; AIES 2023 detail not yet retrieved). 2022: revenue 308bn, expenses 197bn, payroll 90.6bn, depreciation 20.0bn.
BEA IO 2017, 518200	Electricity 1.0% of gross output; intermediates 41%, compensation 19%, gross operating surplus 38% (Section 1.2).
Equinix (10-K, IR)	Revenue \$8.7bn (2024), \$9.2bn (2025); cost of revenues 51%; utilities 28% of cash operating expenses and 11–16% of revenue (about 15% recently), partly passed through.
Digital Realty (10-K)	Utilities expense \$1.0bn (2022), 1.47 (2023), 1.33 (2024), 1.43 (2025) on revenue 4.7–6.1bn: 21–27% of revenue; tenant reimbursements cover 87–94% of it.
Colocation pricing (CBRE H2 2025)	Wholesale asking rent \$196/kW-month (Ashburn > \$215); electricity at 7 cents/kWh and full load is \$51/kW-month, i.e. 21% of the customer's all-in cost or 26% if bundled into rent.
Epoch AI (May 2026), 1 GW AI campus	Capex \$38bn (servers 56–60%, facility \$11.4/W ≈ 30%, network \$4.9/W ≈ 13%). Annual TCO \$8.5bn: electricity \$0.6bn (7%; PUE 1.14, utilisation 71%, 8.3 cents/kWh), other opex \$0.3bn (4%), capital services \$7.6bn (89%), of which servers \$5bn. IT life 5 years, facility 14.
Facility only (shell + mechanical/electrical, no IT)	\$10–12m/MW conventional, \$15–20m/MW AI-ready (Goldman, JLL: shell-and-core \$11.3m/MW). Electricity \$0.5–0.6m/MW-year against capital services \$1.4–1.5m and labour/maintenance \$0.2m: electricity 25–30%, other intermediates ≈ 10%, labour small, capital 60–65%.
McKinsey (April 2025), 2025–30 AI stack	\$5.2tn: 60% chips and hardware, 25% power, cooling and electrical, 15% land, site and shell.

`https://epoch.ai/data-insights/ai-datacenter-cost-breakdown`; `https://www.goldmansachs.com/insights/articles/tracking-trillions-the-assumptions-shaping-scale-of-the-ai-build-out`; `https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/the-cost-of-compute-a-7-trillion-dollar-race-to-scale-data-centers`

3.2 Proposed expenditure shares of the node

The share of electricity depends on what the node's output is. Three consistent definitions:

1. **Computing services, all-in** (output = annual cost of the whole US fleet including IT; the natural definition for a node whose capital includes servers): electricity 7–10% of gross output, other intermediates (telecom, maintenance, software, water) 5–8%, labour 3–5%, capital services 75–85%. Gross output for 50 GW at Epoch's \$8.5m/MW-year for AI and less for conventional capacity is of the order of \$250–350bn in 2025, consistent with SAS 518 revenue (\$375bn in 2024, which includes non-data-center processing).
2. **Facility services** (output = rent of powered shell, IT owned by users, as in colocation): electricity 20–30%, other intermediates 10%, labour 5%, capital 55–65%.
3. **Industry 514 as published**: electricity 1%. Not a data center.

Recommendation: definition 1 with electricity at 8% (sensitivity 5–15%), natural gas 0–2% (rising with on-site generation), and the capital bundle of Section 4. Note that the node's electricity is at most 5% of US electricity: even a doubling of its demand is a 5% shift of the electric power row, so the aggregate effects of the boom on electricity run through the level of investment in generation, not through the intermediate share.

3.3 Depreciation

Hyperscalers extended server lives from 4 to 6 years in 2022–23 (Microsoft, Alphabet, Amazon; Meta to 5.5 in 2025); Amazon moved part of its fleet back to 5 years in 2025, citing AI; Burry (November 2025) argues 2–3 years. Goldman calls the 4–6 year silicon life the most important scale assumption; Epoch annualises at 5 years (3 years raises the annual cost of a GW by 50%). Tax: 5-year MACRS for servers, 39-year for the shell, 100% bonus depreciation after January 2025. BEA (`https://apps.bea.gov/national/pdf/BEA_depreciation_rates.pdf`) has no single geometric rate for computers (Oliner used-price profiles); the 0.3119 often quoted is office and accounting equipment; communication equipment 0.11–0.15; office buildings 0.0247 (life 36). Implied rates in the 5140 detailed accounts: equipment 22%, structures 2.4% (2024). For the node: IT δ 0.25–0.40 (3–6 years), mechanical/electrical 0.08–0.15 (10–20 years), shell 0.025 (36 years); bundle-weighted at the 60/25/15 capex mix this gives $\delta \approx 0.20$–0.28, far above any existing sector.

4 Construction of the node in the 41-sector dataset

4.1 Which sector to split

- **Option A: split `Info` (26) into Data centers and Rest of Info.** Consistent with the classification: hosting, cloud infrastructure and colocation are 518210; hyperscaler own-account capacity is capital of 514/511, both inside `Info`. Data on the 518200 column and row exist for 2007/2012/2017; investment by asset exists annually for 5140. Rest of Info keeps publishing, software, telecom, broadcasting, and internet publishing.
- **Option B: split `Real_estate` (28).** Not supported: BEA records no colocation output in real estate; Equinix and Digital Realty are REITs by tax status only; the real estate sector is dominated by housing services. What real estate does contain is landlords' purchases of electricity for tenants ($41bn in 2023, of which data center landlords are a part) and 15bn a year of office structures investment, some of it data centers built by developers for lease. Treatment: reassign an estimated data center share of ORE's electricity purchases and of its office structures investment to the node.
- **Functional definition.** Follow the PwC/Data Center Coalition and Fed practice and define the node as all US data center capacity (hyperscale, colocation, enterprise) regardless of owner, selling computing services to the sectors that use it, including its former parents. In the model this reassigns capital from `Info`, `Finance`, `Retail`, etc. to the node and adds an intermediate purchase in return; it is an accounting reallocation with no effect on the aggregate.

4.2 Construction by object

Object	Source	Construction	Assumption	Status
Gross output, snapshot 2023/24	FERC 50 GW; LBNL 192 TWh; Epoch/JLL cost per MW; SAS 518 revenue	capacity × annual cost per MW by type (AI vs conventional); cross-check with SAS hosting revenue plus hyperscaler own-use	steady state is a 2023–24 snapshot; no time series	feasible
IO column	2017 detail 518200 column; Epoch/facility cost shares; EIA prices × TWh for electricity	electricity from 2211 at 8% (5–15%); gas from 2212/211 at 0–2%; telecom from Rest of Info; maintenance and software from Prof_Tech and Rest of Info; labour and GOS residual	518200 non-energy structure re-scaled to the engineering shares	feasible
IO row	2017 detail 518200 row; SAS/Census product lines; IDC cloud revenue by customer industry	sales to Rest of Info (hyperscaler own use), Prof_Tech, Finance, Retail, Mgmt, government, PCE (consumer cloud)	drop payment-processing customers; proportional to IT spending by industry	moderate
Investment column (γ^I)	FA 5140 by asset 2019–2024; NIPA 5.4.5U data centers; Census C30	bridge: SOO1, SU20 → Construction; EP1A–H, EP20 → Computers (334); EO72 (UPS, switchgear) → Electrical (335); EI50 (cooling) → Machinery (333); software → Info/Prof_Tech; shares 60/25/15 IT/M&E/shell	vom Lehn–Winberry bridge on 5140 shares; level from NIPA and Fed measure	feasible
Capital stock, δ	FA 5140 net stocks; asset lives above	bundle-weighted $\delta \approx 0.2$–0.28	5140 stock scaled to functional capacity	feasible
Import content	Fed: 80–90% of IT equipment imported	see below	model choice	open
Shock	none of its own	recipient of the boom shock (Section 5)		open
Other sectors	remove reassigned capital and add intermediate purchases from the node	keep 1997–2023 series for the 40 other sectors		feasible

4.3 The import problem

In RbcProdNet the investment good of every sector is produced domestically; sector 9 `Computers` would produce the servers. In the data 80–90% of the high-tech equipment for the buildout is imported (Fed FEDS 2025-109 and Fiori, Lipa and Nuenninghoff 2026), and the Fed's current-account note treats the boom as an investment-specific technology shock that widens the deficit by 0.2% of GDP. Options: (i) keep the closed economy and read the Computers sector's response as world supply; (ii) add an import leak for the IT part of the data center bundle at an exogenous world price, so that the domestic content of the boom is structures, electrical and cooling equipment, and construction; (iii) treat the IT bundle as a non-produced input in fixed supply (a resource constraint, like the fuel rows in `data_notes.tex`). Option (ii) is the honest one and changes the aggregate multiplier of the boom by a factor of about two. The plan adopts (ii) with the import row at an exogenous world price and balanced trade, combined with the US+Nvidia addition to sector 9

(Section 6), so that the domestic content of the IT bundle is 60% gross rather than the Fed's 20%.

5 Modelling the investment boom

5.1 Facts

- **Hyperscaler capex** (cash purchases of property and equipment, $bn; definitions differ across firms, leases excluded unless noted): Amazon 63.6 (2022), 52.7 (2023), 83.0 (2024), 131.8 (2025), about 220 guided for 2026; Alphabet 31.5, 32.3, 52.5, 91.4, 195–205; Meta 31.2, 27.0, 37.3, 69.7, 130–145; Microsoft (fiscal years) 23.9, 28.1, 44.5, 64.6, about 175 calendar 2026 including leases; Oracle 4.5, 8.7, 6.9, 21.2, 55.7 (FY26); CoreWeave 35–39 (2026). Big-4 total about 210–230 (2024), 360–410 (2025), 720–750 guided for 2026. Alphabet: 60% of 2025 capex servers, 40% data centers and networking.
- **Industry totals**: Dell'Oro global data center capex $455bn (2024, +51%), +57% in 2025 (about $715bn), above $1tn in 2026; McKinsey $6.7tn cumulative to 2030 ($5.2tn AI); Goldman $7.6tn over 2026–31 ($765bn in 2026, $1.6tn in 2031; assumes $15m/MW facilities, $2,500/kW new power, 4–6 year silicon life); BIS: AI investment about 1% of GDP in the most exposed economies.
- **GDP accounting**: information processing equipment plus software contributed 1.5pp to annualised real GDP growth in 2025H1 with the rest of GDP at −0.1pp (Furman's "92% of growth"); IPE alone 0.90pp in 2025Q1 and 0.16pp in 2025Q3 (St. Louis Fed); net of imported computers, semiconductors and telecom equipment the AI contribution falls to 0.4–0.5pp (MRB) and Goldman puts the data center boom at 0.2pp of 2025 growth; the Fed's project-based measure gives 0.4pp in 2025 and 0.75–1.1pp in 2026 before imports; Apollo/Slok: data center capex 0.6% of GDP (2023), 1.4% (2025), 3.1% (2027); Bridgewater 1.4pp of growth in 2026. IP and equipment investment as a share of GDP is 0.8pp above 2024 and near its 2000 peak of 11.5% (Li 2026).

5.2 How the literature introduces the boom

Paper	Shock	Result
Fiori, Lipa, Nuenninghoff, FEDS Note 14 July 2026	Investment-specific technology shock (Fisher 2006 SVAR, 1993–2019); the AI boom as its current phase	one s.d.: investment +2%, output +0.5pp, current account −0.2% of GDP; 90% of high-tech equipment imported from East Asia
Li, FEDS Note 6 July 2026 (Li 2007)	Embodied general-purpose technology of unknown size; learning by investing; rational overshooting	railroads, canals, 1990s fiber as analogues; investment can overshoot without irrationality; do not curb investment pre-emptively
IMF WP 25/081 *Power Hungry* (2025)	Exogenous path of data center electricity demand in a global CGE	US electricity prices +8.6% with constrained renewables and transmission; US CO_2 +5.5%
Epoch GATE (Erdil et al. 2025)	Compute stock automates tasks; integrated assessment	compute build-out first, TFP later
Kleinman Center (Penn) 2026	Nodal power-flow GE	a 200 MW data center at a constrained node can double the local electricity price
Acemoglu, *Economic Policy* 2025	Task-based TFP benchmark	TFP +0.66% over ten years (conservative size for a news shock)
Korinek and Suh, NBER w32255 (2024)	Automation of a task continuum	GDP +100–300% over ten years (upper tail)
Beaudry and Portier 2006; Jaimovich and Rebelo 2009	News about future TFP; investment-specific news	the standard way to get an investment boom before measured productivity in a neoclassical model with adjustment costs

No paper introduces the boom as a shock to a sector node in a vom Lehn–Winberry investment network. The timing of 2023–26 (investment first, productivity not yet measured) is that of a *news shock*: an anticipated increase in the productivity of sectors that use computing services, or in the productivity of the data center node itself, announced at t and realised (or not) at $t + k$. This is also the setting in which the planner's precautionary motives from RbcProdNet bite: the news may not materialise, the capital is short-lived and specific, and the electricity it needs is supplied by a sector with long-lived capital. Alternatives: an investment-specific technology shock to the IT bundle (fits 2010–2022 falling GPU prices, not 2024–26 chip and memory inflation); a demand shift toward computing services (rhetorical unless justified by productivity); an exogenous investment path in the node matching the Fed's $370bn by 2026Q2 (useful as accounting and as a robustness check, silent on expectations and on a bust). Existing IO studies of data centers (PwC for the Data Center Coalition, IMPLAN, 2026: 5.5m jobs and $927bn of GDP in 2024; Virginia JLARC 2024: 74,000 jobs, $9.1bn) treat imported IT as domestic value added and are not usable for calibration; JLARC's finding that most local benefits are in construction and that unconstrained Virginia load doubles in ten years is the relevant part.

6 Nowcast inputs for 2025 and the US+Nvidia treatment

The execution plan is `data_center_nowcast_plan.md` in this folder. This section records the facts it relies on, verified on 4 September 2026.

6.1 Vintage of the official network

- Annual IO (71 industries, after redefinitions), GDP by industry and fixed assets run through **2024** (releases of 25–26 September 2025). The 2026 annual update, the first with national, industry and regional accounts on the same day, is **30 September 2026** (revision window

2021Q1–2026Q1; first use of Census AIES 2023; no change to computer service lives; no data-center asset type; no 2022 detailed benchmark announced). Until then 2025 industry accounts are a nowcast. `https://www.bea.gov/information-updates-national-regional-economic-accounts`

- The static zip of all IO tables is no longer served reliably (HTTP 500 on 4 Sept 2026); the 1997–2023 files are saved locally, and 2024 comes from the API (dataset `InputOutput`, integer `TableID`, free key) or from the iTables export. The 138-industry GVC Make/Use/import matrices 2007–2024 (May 2026) are the only official annual tables finer than 71 industries.
- Quarterly GDP by industry: 2026Q1 (25 June 2026). NIPA underlying detail tables 5.4.5U, 5.5.5U and 5.6.5U: 2026Q2 (26 August 2026 second estimate). Census C30: July 2026 (1 Sept 2026). QSS 518: 2026Q2. The vom Lehn–Winberry replication network (Harvard Dataverse, `doi:10.7910/DVN/CALDHX`) stops in 2018; no public extension exists; BEA's last capital-flow table is 1997.

6.2 Nowcast of data-center investment, 2025 and 2026H1

$bn, nominal	2023	2024	2025	2026H1, annual rate
Structures: Census C30 private data centers (annual, latest vintage)	20.0	34.8	49.7	65–75 (May–July SAAR 65.7, 70.8, 75.2)
Computers B935RC (NIPA 5.5.5)	149.1	178.7	270.7	392.4 (Q1 384.2, Q2 400.6)
2015–2022 linear trend ($92.8 + 8.47\,t$)	160.6	169.1	177.6	186.0
residual = Fed "data-center equipment"	−11.6	9.6	93.2	206.4
Communication equipment C275RC	142.9	141.9	163.7	
Fed NIPA method: structures + residual	8	44	143	270–280
Fed Dodge nowcast, mean (FEDS 2025-109, Table 2, Q4 annual rate)	10	58	179	370 (2026Q2); 464 (2026Q4)

The Fed's Table 2 values are fourth-quarter annual rates, not annual totals: a path running from 58 (2024Q4) to 179 (2025Q4) averages about 140 over 2025, consistent with the NIPA-method 143. Annualised, the Fed path is roughly 10 / 40 / 140 / 400bn for 2023–26. The Fed measure is an increment (structures at level, computers in excess of trend) and excludes owner-furnished electrical and mechanical equipment, which is in neither C30 nor the computers residual. 2027 scenarios (Table 2, mean, Q4 annual rate): 357 / 612 / 921bn for plan inflows at 0.25× / 1× / 2× the 2024–25 pace; import adjustment 44% domestic (30% structures, 70% equipment at 80% imported). The 2023 residual is negative on the current vintage: the detrending is only usable from 2024. Communication equipment was flat through 2024 and rose $22bn in 2025, an upper bound for the networking part of the bundle. Hyperscaler cash capex 2025: Amazon 128.3, Alphabet 91.4 (60% servers, 40% data centers and networking), Meta 69.7 (72.2 with finance leases), Microsoft FY25 64.6 (about 88 with finance leases); 2026 guidance 720–745 for the four; Oracle FY26 55.7; CoreWeave 2026 capex 35–39. QSS NAICS 518 revenue 2025: $422bn (upper bound for the node's output; it includes payment processing and non-data-center hosting). Capacity end-2025: about 50 GW online (FERC), 35 GW under construction (JLL). QCEW 518210, 2024: 484,000 jobs at $194,800 average pay, $94bn of wages; CES 518 employment fell from 481k to 473k during 2025 while capex tripled, so the node's labour must not be scaled with its capital.

6.3 Where Nvidia is in the national accounts

- **Nvidia's accounts** (fiscal year 2026 ended 25 January 2026 ≈ calendar 2025; 10-K segment note verified at `https://www.sec.gov/Archives/edgar/data/1045810/000104581026`

000021/R25.htm): revenue $215.9bn (FY2025 130.5, FY2024 60.9); Data Center $193.7bn (compute 162.4, networking 31.4; FY2025 115.2, FY2024 47.5); gaming 16.0. Cost of revenue $62.5bn (gross margin 71.1%), R&D $18.5bn, SG&A $4.6bn, operating income $130.4bn (60.4% of revenue), net income $120.1bn, capex $6.1bn, 42,000 employees. Revenue by customer headquarters: US $149.6bn (69%), Taiwan 42.3, China incl. Hong Kong 19.7, other 4.3; Nvidia estimates that 76% of Data Center revenue from Taiwan-headquartered customers (the ODMs) went to end customers in the US and Europe. The older billing-location series (FY2025: US 61.3, Singapore 23.7, Taiwan 20.6, China 17.1) is an invoicing geography (shipments to Singapore under 2%) and must not be mixed with the HQ series. Long-lived assets: US $5.1bn, Taiwan 3.2, Israel 1.5. Suppliers: TSMC and Samsung (wafers), SK Hynix, Micron, Samsung (HBM), TSMC CoWoS packaging, Foxconn, Wistron, Fabrinet (assembly). GPU manufacturing cost about $6.4k for a B200 (HBM 45%, packaging 17%, logic die under 15%) against a $30–40k price: most of the price is Nvidia's margin. Two direct customers were 22% and 14% of FY2026 revenue.

- **Not in US industry 334.** Industry 334 gross output in the summary Use table: $302bn (2017), 318 (2021), 340 (2022), 342 (2023), while Nvidia's revenue went 27 → 61bn over 2022–23 and is 216bn in 2025. Nvidia's sales are not in the output of computer and electronic products; a fabless firm's establishments are classified by their activity (design: 5415/5417, or wholesale 4234, or headquarters 55), and the ECPC's 2010 recommendation to classify factoryless goods producers in manufacturing was never implemented (2014 Federal Register notice; no FGP industry in NAICS 2027 either).
- **Epoch AI, 24 August 2026**, *The Nvidia-sized hole in US GDP statistics* (Juniewicz, Carey, Trammell, Ho; https://epoch.ai/publications/the-nvidia-sized-hole-in-us-gdp-statistics): US GDP growth understated by about 0.3pp over the past year because the import price of an AI server includes Nvidia's markup, which is never recorded as a US export of goods, IP, services or merchanting (the ODM transforms the chips, so it is not merchanting; Nvidia pays TSMC rather than licensing to it); BEA confirmed to them that change-of-ownership data are not collected. Gap about $30bn a quarter in 2025 (about $140bn a year for semiconductors); could approach 2pp of growth a year by 2028; GDI already contains Nvidia's US wages and profits (about $120bn pre-tax in the year to January 2026), which is one reason GDI and GDP diverge. BPM-7/SNA 2025 would record the sales of a factoryless producer as goods exports; BEA has stated since 2014 that it will not do so in the core accounts.
- **The dispute.** The X exchange between Brian Albrecht and the Epoch authors could not be retrieved. The substantive objections on record (Magica, J.P. Morgan AM commentary): FGP treatment requires that Nvidia (a US resident) be the economic owner of the wafers and chips and bear the production risk, which needs contract-level evidence; a Taiwanese branch with substantial operations would be a Taiwan resident; the 0.3pp has no firm-level reconciliation; BEA's 2014 "no GDP impact" statement concerned a reclassification of recorded flows, not the unrecorded export Epoch describes, so the two are not in contradiction. Related but different measurement points: Korinek and McKelvey (PIIE, May 2026) argue AI output is understated through deflators (inference prices −94% a year), not through trade; Furman, Hatzius, the St. Louis Fed and FEDS Note of 17 July 2026 discuss the import offset to AI investment, not the missing export. For our purposes the dispute matters in one place: Nvidia's operating income is already in NIPA domestic corporate profits (GDI) while the same dollars sit in imports on the expenditure side, so a model economy with "US+Nvidia" has a GDP above official GDP by Nvidia's US-installed value added (about $65–70bn, 0.25% of 2025 GDP, close to Epoch's 0.3pp of growth); we report that and reconcile it against GDI

rather than presenting model GDP as official GDP.

- **Trade side.** US imports of computers (Census end-use 21300) $88.5bn (2023), 116.8 (2024), 218.2 (2025); computers, peripherals and parts (NIPA) 155.7, 217.4, 358.6. Imports from Taiwan $201bn in 2025 (+85bn); Mexico's computer-equipment exports $85bn (95% to the US, 82% HS 8471.50 assembled servers). Onshoring: TSMC Arizona Blackwell wafers in volume since October 2025 but packaged in Taiwan; Wistron Fort Worth opened July 2026 (GB300 boards); Foxconn Houston ramping; the "$500bn of US-made AI infrastructure" is partner output, not Nvidia capex.
- **Implication for the node.** Under the border rule 80% of a data-center IT dollar is imported (Fed). Under US+Nvidia, the accounting is a single identity for the node's 2025 IT purchases (about $200bn: the server share of B935RC plus the networking increment): Nvidia US-installed $100bn (about 52% of data-center revenue, below the 69% customer-HQ share because revenue is recognised at shipment to the ODMs one to two quarters before installation and part of Taiwan-HQ sales is installed abroad), of which 29 is Nvidia's imported cost of revenue, 4 non-US R&D and 67 US value added; other AI-server content $40bn and conventional servers $60bn, each 20% domestic. Domestic content of a data-center IT dollar: 60% gross and about 40% in value added under US+Nvidia, 20% and about 15% under the border rule, 100% in a pure closed economy. The imported part enters as an import row at an exogenous world price financed by a proportional export (balanced trade), not as the fixed-supply device used for fuels. The treatment is Nvidia-only: Broadcom, AMD and Marvell AI silicon, and Google TPU and AWS Trainium (own-account designs that enter US investment at cost, with no market markup) are left as they are. Consistency check: 334's gross operating surplus in the summary Use tables is flat at $94.5bn (2019), 96.4, 99.3, 99.3, 97.5 (2023) while Nvidia's operating income went 2.8 → 4.5 → 10.0 → 4.4 → 33.0bn over the same fiscal years, so Nvidia's profits are not in 334's value added either (through the 2023 vintage); the addition to 334 is therefore a gross addition, to be re-checked on the 2024 tables (FY2025 operating income 81bn).

7 Hard steps and open items

1. **Level of the node.** No official gross output or capital stock for data centers as a function. The level comes from capacity (FERC 50 GW; LBNL TWh) times engineering cost; SAS 518 revenue and the Digital Economy cloud series are cross-checks. Document the sensitivity of Domar weight and upstreamness to this choice.
2. **Row re-weighting.** The 518200 row is payment processing and government; cloud revenue by customer industry (IDC, Synergy) is commercial. Alternative: proportional to each sector's investment in computers and software (FA detail), which is official.
3. **Import content of the bundle** (Section 4): a model decision, not a data one; it halves the domestic multiplier.
4. **Shock and its size**: news about future productivity of which sectors, of what size (Acemoglu 0.66% to Korinek–Suh), with what probability of realisation. Interacts with the P&P length.
5. **Snapshot steady state**: the node's share of investment goes from 1% of GDP (2023) to 3% (2027 projections); the steady state is a year, and the boom is a transition. The energy split of `data_notes.tex` has the same issue with renewables.
6. **Electricity interaction**: the node's marginal electricity is gas-heavy (Virginia, Ohio, Texas) and part of it will be on-site gas; requires the electricity split into fossil and non-fossil generation to matter. The channel from the boom to gas demand is investment in generation

(EI21 steam and EI22 internal combustion engines, gas turbines in Machinery/Electrical) and utilities' capex response, not the intermediate share.

7. **To retrieve**: NIPA 5.4.5U data center values 2020–2026Q2 (BEA interactive tables; not on FRED); Census C30 historical Excel for 2014–2018 and a consistent monthly 2025–26 path on the September 2026 vintage (`privsatime.xlsx`); 2024 Use table and GDP by industry (BEA API key); AIES 2023 operating expenses for 518 (electricity line, if published) and the AIES 2023 capex experimental tables (April 2026, 3–4 digit NAICS); Economic Census 2022 product lines for 518210 (colocation, cloud); Digital Economy 2017–2022 cloud spreadsheet. Retrieved: QCEW 518210 (2024 annual), FRED NIPA series, Nvidia 10-K, Census C30 annual by region.
8. **Not available**: TFP or output series for data centers; hyperscaler power costs in 10-Ks; official share of on-site gas in data center energy; US capacity by owner type (Synergy US split is paid).